

Cardiorespiratory Sound Separation Using Singular Spectrum Analysis

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Abstract—Cardiorespiratory signal separation is a critical task in biomedical signal processing, with significant implications for improving the accuracy of cardiovascular and pulmonary disease diagnoses. The presence of complex mixed signals and noise complicates the separation of cardiac and respiratory sounds from a single-channel recording. Traditional separation methods struggle due to the overlapping frequency ranges of these physiological signals. In this work, we propose a multi-stage Singular Spectrum Analysis (MSSA)-based framework for the separation of cardiac and respiratory sounds. The approach utilizes a two-stage SSA process where the first stage isolates the cardiac components through periodic structure analysis, while the second stage extracts respiratory components based on relative variance and cross-correlation. Key features of this method include a systematic selection of RCs that ensure high-quality signal separation, preserving physiological integrity. The framework was validated on real-world cardiorespiratory recordings obtained via electronic digital stethoscopes, demonstrating robustness across varying recording conditions.

Keywords—Biomedical signal processing, Cardiorespiratory sound separation, Heart and lung sounds, Multi-stage Singular Spectrum Analysis

I. INTRODUCTION

Cardiorespiratory separation is a challenging task in biomedical signal processing, enabling more precise diagnosis of cardiovascular and pulmonary diseases [1]. During inhalation and exhalation, air moves through the trachea and branching airways which create normal respiratory sounds due to the turbulence produced as it flows through the respiratory passages [2]. Cardiorespiratory sounds can be detected through the chest wall using a stethoscope or captured with microphones placed on the surface of the chest [3]. Separating cardiac (C) and respiratory (R) sounds helps clinicians detect diseases of each system more reliably. For example, respiratory disease diagnosis like pneumonia and COPD can benefit from cardiac sound cancellation. Abnormal respiratory sounds like crackles or wheezes are easier to identify when cardiac beat is filtered out [4]. In addition, cardiac sound analysis plays a crucial role in detecting cardiovascular diseases (CVDs) which are among the leading causes of death worldwide [5]. The ability to separate cardiac and respiratory sounds has far-reaching applications in biomedicine, from improving disease diagnostics and wearable monitoring to enabling advanced AI auscultation and new telehealth solutions. Those applications are contributing to

better healthcare outcomes. However, due to the complexity of the mixed signals and the presence of noise, achieving high-quality separation from a single-channel signal remains a challenging task.

Numerous blind source separation (BSS) techniques have been proposed to address this challenge. Traditional methods include adaptive signal processing (ASP) approaches such as the least mean square (LMS) algorithm and adaptive noise cancellation (ANC) techniques. These methods require a reference signal which is often unavailable in practical clinical settings [6][7]. Similarly, independent component analysis (ICA) has been widely applied for multivariate source separation [8], yet it fails in single-channel scenarios due to its reliance on multiple inputs. Nonnegative matrix factorization (NMF) has gained popularity due to its ability to separate mixed signals without prior information. However, standard NMF struggles with spectral overlap that reducing separation efficiency [9]. Recent advancements have introduced multi-constrained NMF (MCNMF) which incorporates sparsity and smoothness constraints to improve decomposition accuracy [9]. Deep learning-based methods, by integrating deep autoencoders (DAE) with NMF and variational mode decomposition (VMD), have demonstrated superior performance in extracting meaningful signal features and enhancing noise robustness [10]. Another notable approach is the embedding space centroid network (ECNet) which combines convolutional neural networks (CNN) for cardiac sound classification with bidirectional long short-term memory (Bi-LSTM) networks for signal separation. It results in great performance in terms of signal-to-distortion ratio (SDR) and signal-to-noise ratio (SNR). Meanwhile, transformer-based architectures such as the Audio Spectrogram Transformer (AST) have been applied to respiratory sound classification. This is a purely attention-based model pretrained on visual and audio datasets in improving the classification Score of respiratory. The proposed Patch-Mix augmentation randomly replaces patches with those from different samples. The method breaks records on ICBHI dataset and outperforms all exiting method [11].

Singular Spectrum Analysis (SSA) has gained attention due to its effectiveness in decomposing time-series data into interpretable components without prior information. SSA operates by embedding a signal into a high-dimensional trajectory matrix, performing eigen decomposition and

reconstructing components based on spectral energy contributions [12][13][14]. SSA has emerged as a powerful data-driven approach for signal decomposition and extracting meaningful subcomponents from time-series data, especially in biomedical signal processing tasks such as noise reduction and signal extraction. A previous study was focused on the problem of cardiac sound interference in respiratory sound (RS) recordings and presents Singular Spectrum Analysis (SSA) as a method to localize primary cardiac sound components within mixed signals. Due to the frequency overlap between cardiac and respiratory sounds, traditional filtering methods are insufficient. Instead, SSA's eigenvalue spectrum analysis was used to distinguish components related to cardiac sounds [15]. Another study proposes an SSA-discrete cosine transform (DCT) method for respiratory sound denoising. It tackles the issue of ambient noise contamination in respiratory sound recordings by first applying SSA to decompose the signal and then using DCT to attenuate noise while preserving respiratory information. The results demonstrate higher SNR improvements compared to wavelet-based methods, making SSA-DCT a promising approach for respiratory signal improvement [12].

In this study, a multi-stage SSA-based framework (MSSA) for Cardiorespiratory separation was proposed. Its distinctive feature is the multi-stage decomposition process where the first SSA stage extracts dominant periodic components associated with cardiac activity, while a secondary SSA stage further isolates respiratory components based on correlation within RCs. A key aspect of the method is the systematic selection of Reconstructed Components (RCs). This ensures that cardiac-related components with strong periodic structures are preserved, while respiratory-related components are extracted based on their energy contribution in the spectral domain. The algorithm was tested on random real Cardiorespiratory recordings obtained using electronic digital stethoscopes, demonstrating its robustness across different recording conditions. The advantages and limitations of the proposed method are discussed, with its primary limitation being the difficulty in fully separating overlapping broadband components that share similar frequency characteristics. However, compared to conventional filtering and matrix factorization techniques, the proposed MSSA framework preserves physiological signal integrity while reducing interference, making it a promising approach for biomedical signal processing applications.

To evaluate the research approach. Key performance metrics such as signal-to-distortion ratio (SDR), short-time objective intelligibility (STOI) and ground truth correlation are computed to compare the proposed method against existing techniques.

II. METHODOLOGY

A. Basic Algorithm

The SSA procedure consists of four main steps: Embedding, Singular Value Decomposition (SVD), Grouping and Reconstruction [16][17]. In this method, SSA is performed twice for further refinement.

The SSA decomposition was applied to raw audio signals to extract dominant frequency components corresponding to

Cardiorespiratory sounds. This process consists of the following steps.

- **Trajectory Matrix Construction:** A time-delayed embedding method is employed to create a trajectory matrix of the signal using a predefined window length L .
- **SVD:** The trajectory matrix is decomposed into singular values and singular vectors, capturing the principal modes of variability in the signal.
- **Grouping and Reconstruction:** The decomposed components are analysed to separate meaningful components corresponding to cardiac and respiratory sounds. These components are then reconstructed into separate time-domain signals.

A key factor in SSA performance is the selection of the window length which influences the accuracy of extracted components. An optimal selection of window length for SSA-based blind source separation was explored, demonstrating its impact on decomposition effectiveness [17]. However, previous research often employs longer window lengths to improve accuracy, but it significantly increases computational complexity. In this work, a window length of 50 was selected to achieve a balance between processing speed and signal separability to secure efficient and rapid biomedical signal analysis. It allows for faster computations while maintaining sufficient spectral resolution.

Given an audio signal $X = x_1, x_2, \dots, x_T$ of length T , a trajectory matrix X is constructed using window length L :

$$X = \begin{bmatrix} x_1 & x_2 & \cdots & x_K \\ x_2 & x_3 & \cdots & x_{K+1} \\ \vdots & \vdots & \ddots & \vdots \\ x_L & x_{L+1} & \cdots & x_T \end{bmatrix} \quad (1)$$

Where $K = T - L + 1$ is the number of lagged vectors. This transformation captures temporal dependencies in the signal. Then SVD is applied to X to obtain the singular values and vectors:

$$X = \sum_{i=1}^d \sqrt{\lambda_i} U_i V_i^T \quad (2)$$

Where λ_i are the eigenvalues of XX^T and U_i, V_i are the singular vectors. The largest singular values correspond to the dominant signal components, while smaller values represent noise.

B. Proposed Method

The flowchart in Fig. 1 shows the MSSA framework designed for Cardiorespiratory separation. The process begins with the input signal which undergoes SSA decomposition with a selected window length of 50. SSA decomposes the signal using embedding and SVD, followed by reconstruction through diagonal averaging and grouping.

The power spectral density (PSD) analyses the frequency content of the reconstructed signals to ensure proper separation.

$$P_{xx}(f) = \frac{1}{N} \sum_{n=1}^N |X_n(f)|^2 \quad (3)$$

Where $X_n(f)$ is the Fourier transform of each windowed segment n . N is the total number of overlapping segments.

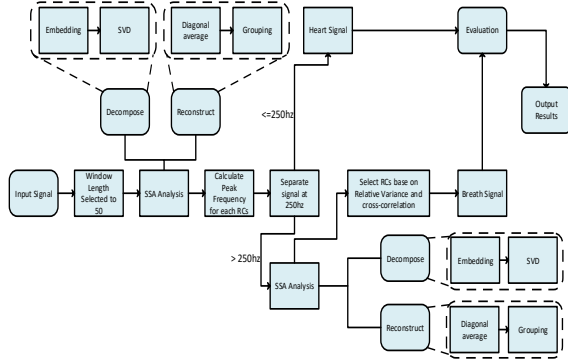


Fig. 1. Flow chart of MSSA.

The next step involves a 250 Hz frequency-based separation criterion, where components with dominant frequencies below or equal to 250 Hz are classified as cardiac-related signals, while components with frequencies above 250 Hz undergo a second-stage SSA for further processing. The separation threshold of 250 Hz was selected based on the typical frequency ranges of cardiac sounds. The first cardiac sound (S1) primarily occupies 100–200 Hz, while the second cardiac sound (S2) extends up to 250 Hz. Given that respiratory sounds predominantly contain higher-frequency components beyond this range, this cutoff secure effective cardiac sound preservation while enabling further SSA-based decomposition of respiratory signals. This choice aligns with clinical auscultation practices, where cardiac sound analysis typically focuses on this frequency range for diagnostic purposes [18]. After the initial separation focusing on cardiac signals, the second SSA step is applied specifically to enhance respiratory signal extraction.

In this secondary SSA, eigenvalues are normalised relative to their total sum to quantify each mode's significance. Given that the SSA decomposition produces 50 RC layers, an average contribution per mode is calculated as 2% (0.02). Hence, a threshold of 2% is set to identify primary modes contributing significantly to the respiratory signal. Subsequently, a cross-correlation analysis is conducted between the identified high-energy modes and the remaining modes, modes showing a correlation above 50% are additionally included, effectively preserving critical respiratory characteristics while filtering out irrelevant or noisy components. This method effectively balances robustness and accuracy, significantly enhancing respiratory signal clarity and reliability in practical respiratory sound extraction tasks.

The Fig. 2 presents the results of SSA applied to respiratory signals, displaying the relationship between relative variance contribution and correlation coefficients across principal components and ground truth signal. As expected, several initial components show both high relative variance and strong

correlation, indicating their critical role in capturing the main structure of the respiratory sound. However, the Fig. 2 also reveals that certain components with relatively low variance still exhibit notably high correlation with the ground truth, suggesting that variance alone may not fully capture all signal components essential for accurate reconstruction. To achieve an optimal respiratory signal reconstruction, it is necessary to perform additional cross-correlation analysis between the initially selected RCs and the remaining RCs.

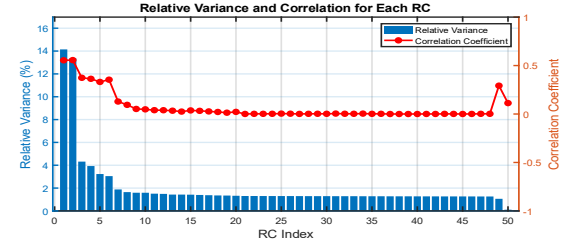


Fig. 2. Relative Variance and correlation within RCs.

Fig. 3 illustrates an example of the proposed MSSA applied to a single-channel chest auscultation recording, clearly showing how cardiac and respiratory sounds are separated. The first stage reconstructed cardiac sound closely matches the ground truth cardiac waveform, the characteristic cardiac sound peaks and periodic patterns align almost perfectly in timing and amplitude. This indicates that the first SSA stage successfully extracted the cardiac component, capturing the periodic S1/S2 cardiac sound cycles while effectively suppressing the overlapping respiratory noise in the mixture. The bottom row of Fig. 3 shows the Stage-2 MSSA results for the respiratory component. The reconstructed respiratory waveform closely follows the ground truth breathing pattern, preserving the continuous, flow-like oscillations of the respiratory sound while effectively removing the previously prominent heartbeats and noise. The timing and envelope of each respiratory cycle in Stage-2 MSSA output align with those of the ground truth respiratory sound, demonstrating that the second SSA stage effectively extracted the respiratory component once the cardiac sound was removed. Overall, this example results highlights the efficacy of the MSSA framework. The excellent visual alignment between each reconstructed signal and its corresponding ground truth waveform confirms that the method preserves the periodicity of the cardiac sounds and the continuous nature of the respiratory sounds.

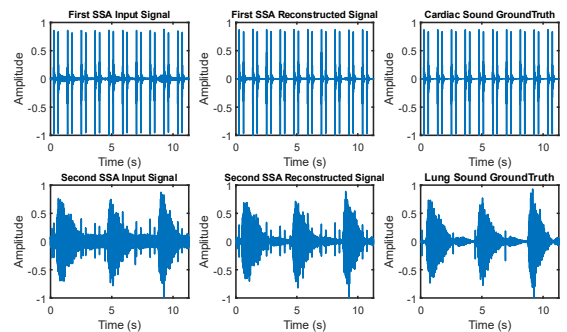


Fig. 3. Reconstruction signal by selected RCs.

C. Evaluation Metrics

To assess the quality of extracted RCs, several evaluation metrics are employed. Signal-to-Distortion Ratio (SDR) measures the ratio of the desired signal to the distortion. Short-Time Objective Intelligibility (STOI) evaluates the clarity and intelligibility of separated signals. Correlation quantifies the similarity between extracted signals and ground truth references provides a measure of reconstruction accuracy.

III. RESULTS AND DISCUSSION

A dataset of combined cardiac and respiratory sound signals was constructed for analysis. The dataset was generated by systematically merging individual cardiac and respiratory sound recordings. The noise was added to the signal and through a normalisation process. Cardiac and respiratory sound recordings were sourced from public dataset [19]. The dataset included a diverse set of normal recordings which were representative of different physiological conditions. To ensure uniformity across the dataset, any multi-channel recordings were converted to single channel by averaging across channels. Additionally, to handle variations in sampling rates, all signals were resampled to the lowest sampling rate found in the dataset. This approach was to make sure of consistency in further processing steps.

A combinatorial approach was used to systematically merge the cardiac and respiratory sound signals. Every cardiac sound recording was paired with every respiratory sound recording, generating a diverse set of mixed signals. Additionally, Gaussian noise was introduced to the combined signal with the noise level set as a fraction of the root mean square (RMS) amplitude of the signal [20]. The value was set to 0.02 which means that the added noise intensity is 2% of the RMS value of the combined signal. The research uses 10 random select cardiac and 5 random select respiratory to generate 50 combined data for testing.

The performance of the proposed MSSA framework was thoroughly evaluated and compared against existing methods, including first stage SSA, HPSS, NMF, Butterworth filtering, and NMF combined with an Autoencoder. TABLE I summarises the evaluation metrics, including SDR, STOI and Correlation for both cardiac sound and respiratory sound components.

TABLE I. MSSA PERFORMANCE RESULTS WITH BENCHMARK

Method	SDR (dB)		STOI		Correlation (%)	
	C	R	C	R	C	R
First stage SSA	26.4	4.6	0.4	0.6	99.2	77.2
MSSA	26.4	5.3	0.4	0.6	99.2	80.5
HPSS	2.5	-22.0	0.2	0.5	60.6	10.1
NMF	1.0	-2.3	0.6	0.2	50.0	10.2
Butterworth Filter	5.7	-5.7	0.5	0.6	91.4	29.7
NMF and Autocoder	4.5	-17.3	0.5	0.6	90.7	31.2

The evaluation results of the MSSA are presented, followed by a comparative analysis with first stage SSA, HPSS, NMF, Butterworth filtering and NMF with Autoencoder. The performance of each method is assessed based on SDR, STOI and Correlation (%) for both cardiac sounds and respiratory sounds. The objective is to determine whether the additional SSA stage improves the separation quality of respiratory sounds while preserving the fidelity of cardiac sounds.

The first observation from the TABLE I is that SSA-based methods achieve significantly higher SDR values compared to the other techniques. The proposed MSSA achieves an SDR of 5.34 dB for respiratory sounds, improving upon the first stage SSA (4.64 dB), while maintaining the same high SDR (26.44 dB) for cardiac sounds. This indicates that the second SSA step improved the separation of respiratory sounds without introducing distortions into the cardiac sound signal. In contrast, HPSS and NMF-based methods exhibit significantly lower SDR values for both cardiac and respiratory sounds, with HPSS yielding 2.50 dB for cardiac sounds and negative SDR values for respiratory sounds, highlighting severe signal degradation. Similarly, Butterworth filtering and NMF with Autoencoder produce suboptimal results, with Butterworth filtering achieving only 5.60 dB for cardiac sounds and negative SDR values for respiratory sounds, further confirming that simple filtering methods are insufficient for effective cardiac-respiratory sound separation.

A similar trend is observed in STOI scores, where the MSSA maintains 0.40 for cardiac sounds and 0.59 for respiratory sounds correspond to the first stage SSA method, confirming that the additional SSA step does not degrade perceptual intelligibility. Other methods, particularly NMF and HPSS, show lower STOI values, with HPSS yielding 0.21 for cardiac sounds and 0.57 for respiratory sounds, indicating a loss of signal clarity. Despite achieving a relatively high respiratory sound STOI in Butterworth filtering, it still performs poorly in terms of SDR which suggesting that it cause significant distortion, removing or distorting both noise and essential respiratory sound components.

The correlation analysis further validates the effectiveness of SSA-based approaches. The MSSA achieves a respiratory sound correlation of 80.47%, demonstrating that the second SSA stage better preserves the respiratory signal structure compared to 77.20% for first stage SSA. For cardiac sounds, both SSA-based methods achieve nearly perfect correlation, confirming that SSA effectively isolates cardiac sounds while maintaining their original characteristics. In contrast, HPSS and NMF produce very low correlation values for both cardiac and respiratory sounds, with HPSS yielding only 60.57% for cardiac sounds and 10.17% for respiratory sounds, reinforcing their limitations in biomedical signal separation.

Overall, the results confirm that the MSSA outperforms traditional filtering, source separation and deep learning-based methods. The additional SSA stage improves respiratory sound separation without affecting cardiac sound integrity, making it a highly effective approach for biomedical signal processing applications. The ability of SSA to leverage both spectral characteristics and eigenvalue-based selection ensures robust

decomposition, even in challenging scenarios with overlapping cardiac and respiratory sounds.

IV. CONCLUSION

In this study, a MSSA framework was proposed and successfully applied for the separation of cardiac and respiratory sounds in single-channel auscultation recordings. By leveraging SSA's capability to decompose complex signals into meaningful RCs, the method effectively separated respiratory and cardiac signals. The key achievements of this research include significantly improved respiratory sound clarity, increased SDR compared to conventional methods and maintaining high correlation levels for cardiac sounds. The results demonstrated that the MSSA consistently outperformed other existing approaches such as HPSS, NMF, Butterworth filtering and NMF combined with an Autoencoder, confirming its robustness and suitability for biomedical signal processing. Future improvements could focus on optimising window length selection, adaptive parameter tuning, RCs analysis and integrating machine learning techniques to reduce computational demands further and enhance the separation accuracy to extend the method's practical application in clinical diagnostic settings.

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